

Yeruru Asrar Ahmed

| Lead Engineer | Generative AI | Computer Vision

+91 98823 65746 • ✉ asrar@cse.iitm.ac.in, asrarabrar21@gmail.com

🌐 www.linkedin.com/in/yeruru-asrar-ahmed-63670113b

github.com/asrarabrar21 | Google Scholar

Professional Summary

Lead Engineer at **Samsung Research Institute Bangalore**, working on AI for personalization and computer vision. Ph.D. from **IIT Madras** with research in **text-to-image generation**, representation learning, and segmentation. Interested in practical AI systems that connect research ideas with reliable product experiences.

Core Research and Technical Interests

Generative AI, Computer Vision, Text-to-Image Generation, Representation Learning, Segmentation, Vision-Language Models, Federated Learning, Edge AI, and AI Systems.

Professional Experience

- Samsung Research Institute Bangalore

Lead Engineer

May 2025–Present

 - **Platform Intelligence Team:** Working on AI features for Samsung devices, including personalization, action prediction, and privacy-aware learning.
 - **Nudge Action:** Built and trained an action prediction model for mobile personalization workflows.
 - **Personalized Data Engine:** Modelled system usage patterns to support smarter background task scheduling.
 - **Personalization Research:** Studied ways to personalize image models across devices while handling differences between users.
 - **Efficiency:** Explored lighter model adaptation methods for practical deployment on devices.
- 8bit.ai (CtrlS–Cloud4C)

AI Scientist

Dec 2024–Mar 2025

 - Built AI and computer vision tools for enterprise analytics, safety monitoring, and compliance tracking.
 - Collaborated with manufacturing and finance clients on deployable AI solutions.
 - Worked on enterprise AI workflows for the Neutrino analytics platform.
- Sony Research India

Research Intern

Nov 2023–Aug 2024

 - Researched faster text-to-video generation while keeping visual quality and text alignment strong.
 - Used efficient training methods and caption refinement to improve video generation workflows.

Publications

- **Unsupervised Co-generation of Foreground–Background Segmentation from Text-to-Image Synthesis**, WACV 2024; extended version in CVIU 2024.
- **End-to-end Training for Text-to-Image Synthesis using Dual-Text Embeddings**, TMLR 2025.
- **Self-Distillation on Conditional Spatial Activation Maps for FG–BG Segmentation**, CVPRW 2024.
- **Breast Cancer Segmentation using UNet and Global Convolutional Networks**, ICPR 2024.
- **Twitter Sentiment Visualization using Deep Learning**, ICIoTCT 2020.

Education

Indian Institute of Technology Madras	CGPA: 8.5/10
Ph.D. in Computer Science and Engineering	2018–2025
Thesis: <i>Text-to-Image Synthesis with Novel Representations and Corollaries</i>	
National Institute of Technology Hamirpur	CGPA: 9.42/10
M.Tech in Computer Science	2016–2018
GITAM University, Visakhapatnam	CGPA: 8.5/10
B.Tech in Computer Science	2012–2016

Selected Research Projects

- **COS-GAN:** Co-generation of images and segmentation masks using spatial co-attention; demonstrated unsupervised learning for image synthesis and segmentation.
- **Dual-Text Embedding GAN:** Jointly optimized multi-level text conditioning for semantically accurate image generation.
- **Self-Distillation Maps:** Foreground-background segmentation via single-step distillation and reconstruction networks.
- **Residual Multi-Scale CNN:** Dense atrous convolutional model for semantic segmentation on PASCAL VOC.

Academic Service and Teaching

- Reviewer for **CVPR** and **ECCV**.
- Teaching Assistant at IIT Madras (2018–2023) for courses including Deep Learning, Pattern Recognition and Machine Learning, Operating Systems, and Computer Architecture.
- Mentored 20+ undergraduate projects in computer vision, GAN-based synthesis, and deep learning systems.

Awards and Achievements

- **Samsung New Vallery Project Award.**
- **Samsung Part of Team Awesome Award.**
- **Samsung AI Expert** – Workera.
- **Star Teaching Assistant** Award (3x) – Department of Computer Science, IIT Madras.
- Secured **GATE Rank 2674 (2017)** and **3612 (2016)** among 100,000+ candidates.
- Represented university in **Inter-University Football Tournament, South Zone, Calicut University, 2014.**

Technical Skills

- **Programming:** Python, C, C++
- **Frameworks:** PyTorch, TensorFlow, OpenCV, HuggingFace Transformers
- **Expertise:** Diffusion Models, GANs, Vision-Language Models, Federated Learning, Edge AI, LLM Integration

Additional Information

- **Hobbies:** Football, Cooking, Movies
- **Languages:** English, Hindi, Telugu

Declaration

I hereby declare that the information furnished above is true to the best of my knowledge and belief.

Place: Bengaluru, Karnataka

Yeruru Asrar Ahmed

Date: March 1, 2026